

AI-Assisted Emergency Department Triage Pilot

Mercer General Hospital — Final Deliverable

1. Problem Statement

Triage nurses at Mercer General Hospital's Emergency Department make high-stakes prioritisation decisions under time pressure, and disagreement between assessors widens during busy periods. Patients with serious but initially unremarkable presentations risk under-triage; lower-acuity cases risk consuming capacity ahead of them. Published AI-assisted triage tools report strong algorithmic accuracy, but none has been validated against Mercer's patients, workflows, or data. This 12-week pilot tests, using de-identified Mercer ED records, whether a locally trained AI decision-support tool can achieve at least 80% agreement with nurse-led acuity judgements and reduce inter-assessor disagreement by at least 20%.

2. Proposed Solution

The proposed system is an AI-assisted clinical decision-support tool designed to augment, not replace, nurse-led triage. Machine-learning models algorithms that identify patterns in historical data to generate predictions will be trained exclusively on de-identified Mercer ED records, directly addressing the literature's most consistent weakness: that published systems have not been validated against local, institution-specific data [6, 8, 9]. Before any future deployment beyond Mercer General Hospital, the model will undergo prospective external validation using site-specific datasets to mitigate the training-distribution shift identified as Risk 1 in the accompanying risk register. This recognises that strong performance within one institution does not guarantee equivalent performance in different patient populations or clinical environments.

The system accepts presenting complaints, vital signs, and available demographic data as inputs and returns a recommended acuity level alongside a confidence score and the key clinical factors driving that recommendation. Presenting the principal factors influencing each recommendation allows clinicians to understand why the model reached a particular conclusion rather than treating the output as a "black box". Nurses retain full authority to scrutinise, accept, or override every recommendation. This human-in-the-loop approach directly addresses Risk 5 (Automation Bias) by preserving independent clinical judgement and ensuring that AI functions solely as decision support rather than a replacement for professional assessment.

Three workflow constraints shape the design directly. To address the Vital Sign Documentation Lag, the system generates recommendations only when sufficient clinical information is available and clearly flags incomplete data so that clinicians understand the limitations of the output, reducing the likelihood of incomplete-record errors associated with Risk 2. To address Nurse Cognitive Load, recommendations appear as a passive overlay within the existing triage interface without introducing mandatory interaction steps, helping minimise workflow disruption while reducing the likelihood of alert fatigue (Risk 3). To address External Review Dependency, each

recommendation is timestamped and the system prompts reassessment whenever new laboratory results, imaging findings, or specialist reviews materially change the available clinical information, reducing the risk of outdated recommendations influencing ongoing care.

Performance will be evaluated against agreement with historical triage decisions, sensitivity for high-acuity cases, reduction in prioritisation variability, clinician-rated usability, and appropriate override behaviour. Bias audits of the training data, as advocated by Chin et al. [15] and Labkoff et al. [14], will be conducted before model training to identify and address differential recording patterns across demographic groups. Performance will also be monitored across relevant patient subgroups to detect inequitable outcomes, supporting mitigation of the equity risks identified in the accompanying risk register.

While these safeguards strengthen the proposed system, they do not eliminate uncertainty. External validation substantially improves confidence in generalisability but cannot guarantee equivalent performance across every healthcare setting. Any future deployment beyond Mercer General Hospital will therefore require prospective evaluation, ongoing clinical governance, and continuous monitoring before broader implementation can be considered safe.

3. Literature Review

The following review is organised into five thematic areas.

3.1 Algorithmic Performance: What AI Can Do

The starting point for any honest appraisal of AI-assisted triage is that machine-learning models demonstrate strong performance when tested on the same data used to train them. The more instructive question — what that performance means for a nurse standing in front of a patient — receives far less attention in the literature.

Bhuvaneshwari et al. [1] built a hybrid framework that combined ensemble methods (where multiple decision models vote together to produce a more reliable output than any single model) with an explainability layer that surfaced the clinical reasoning behind each recommendation. Working in resource-constrained settings where triage inconsistency carries particular risk, they reported high accuracy and argued that making the reasoning visible is essential to earning clinician trust — not optional. They acknowledged, however, that the system had never been run prospectively and that the question of whether nurses change their decisions when the tool flags a disagreement remained unanswered.

Porto [8] synthesised evidence across multiple departments on whether feeding free-text nursing notes into models alongside structured vital-sign inputs improved prediction. It did, consistently: models that drew on both data types outperformed those relying on structured inputs alone. Porto was candid about two methodological limitations — reporting standards varied so widely across included studies that direct numerical comparison was unreliable, and not one study had validated its model at a second institution. Every accuracy figure in that review was, in effect, self-reported.

Wicha et al. [4] illustrated why that matters. Training a gradient boosting model — one that builds predictions by iteratively correcting prior errors — to classify Emergency Severity Index (ESI) levels in febrile patients with a rapid heart rate at a Thai tertiary hospital, the model performed well within that target population. The authors were candid that a single-site, single-demographic design tells you something useful and narrow: that the model works for those patients, in that hospital. It tells you almost nothing about anywhere else.

3.2 Implementation Barriers: Why Performance Alone Is Not Enough

If predictive accuracy were the only barrier, AI triage would already be standard practice. A parallel strand of research examines why clinicians who acknowledge a tool performs well still choose not to use it.

Da'Costa et al. [2] conducted a narrative review of benefits and challenges across EDs and found a consistent pattern: resistance centred on opacity rather than accuracy. When a system assigned an acuity level without explaining the clinical signals behind it, nurses disengaged. The review identified clinician trust, algorithmic bias across age and ethnicity, and the absence of clear override governance as the primary structural barriers. A striking finding was that most reviewed studies never collected data on whether clinicians followed the AI recommendations — the question of real-world impact had been left largely unasked.

A systematic review of AI in emergency and disaster triage [3] reinforced this from a different angle. Sensitivity for high-acuity cases was generally strong, but implementation attempts encountered infrastructure gaps, patchy data quality, and a near-universal absence of governance frameworks for the moment when a nurse and the algorithm disagreed. Wide methodological variation across included studies made firm conclusions about deployment strategies difficult to draw.

The Cureus scoping review [6] added a system-level observation: AI tools that improved throughput in simulation consistently encountered unexpected workflow friction in live deployment. All institutions in that review shared a profile — high-income, technologically mature, well-resourced — leaving their findings of uncertain relevance to hospitals operating under different constraints.

3.3 Ethics, Safety, and Equity: The Non-Technical Requirements

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Evidence is accumulating that the critical barriers to responsible AI triage are not algorithmic but structural — rooted in the data, governance, and deployment decisions made by institutions rather than in the models themselves. Four papers in this review address these obligations directly.

Chin et al. [15], reporting findings from an Agency for Healthcare Research and Quality (AHRQ) Technical Expert Panel published in JAMA Network Open, established a conceptual framework and guiding principles for detecting, mitigating, and preventing algorithmic bias across the healthcare algorithm lifecycle. The panel examined bias across all clinical uses — diagnosis, prognosis, risk stratification, treatment, and resource allocation — covering precisely the domains ED triage occupies. A central argument is that algorithmic bias is neither inevitable nor merely technical: it results from conscious decisions by developers, deployers, and regulators, and can equally be prevented by conscious countermeasures. The framework requires multi-stakeholder accountability structures encompassing clinicians, institutions, developers, payers, and

regulators, and it proposes audit processes for training-data composition, model validation, and post-deployment monitoring as distinct institutional responsibilities. The limitation the panel acknowledged is that adoption is voluntary: without regulatory mandate, the framework creates no binding obligations.

Chang et al. [13] provided the empirical mechanism that the Chin et al. framework addresses at the principled level. In a retrospective matched cohort study across two US teaching hospitals, the authors paired Black and White adult ED patients on age, biological sex, chief complaint, and triage score, then measured differential rates of laboratory test ordering. Even after matching for acuity, White patients received higher rates of blood panel and metabolic testing at both sites. Because machine-learning models treat differential ordering patterns as informative signal, those disparities are encoded into model weights during training and reproduced in subsequent predictions. For any institution building a model on historical records — as this pilot proposes — the implication is direct: without a training-data bias audit, the resulting system may reproduce the inequities it is supposed to be neutral about. The authors noted that the binary racial categorisation used obscures variation within and across demographic groups, and that the retrospective design of a single country limits the generalisability of the finding.

Hinson et al. [12] examined equity outcomes in a multisite live deployment of an AI-assisted triage system across academic emergency departments. The study found that equity-relevant performance improved following AI implementation but that site-specific model calibration was not optional: a model transferred without recalibration performed materially worse on both accuracy and equity measures. This convergence with Chang et al. [13] reinforces a consistent point from two independent study designs — differential historical patterns of care at any institution are encoded in its training data, and institution-specific calibration is a prerequisite for equitable performance, not an enhancement. The study's limitations include a short post-implementation observation window and the academic-centre profile of all participating sites.

Labkoff et al. [14] addressed the patient safety and governance dimension through a structured multi-stakeholder consensus process convened by the American Medical Informatics Association (AMIA) and Beth Israel Deaconess Medical Center, involving over 200 participants including clinicians, developers, ethicists, lawyers, policymakers, and patients. The working group identified five areas of governance failure in current AI clinical decision-support deployment: insufficient safety assurance during development; absence of standardised validation and certification; no national safety-monitoring or incident-reporting infrastructure; insufficient transparency of AI outputs to clinicians; and inadequate human-oversight mechanisms. The group recommended that patient safety does not emerge passively from technical performance and that each governance gap requires deliberate institutional and regulatory action. The inherent limitation is that consensus statements carry no binding force: implementation depends on voluntary institutional commitment.

Together, references [12], [13], [14], and [15] establish that the ethical obligations of AI triage deployment are distinct from, and not reducible to, algorithmic performance metrics. A system can be accurate on average while encoding structural inequities, operating without meaningful oversight, or imposing unjustifiable risks on specific patient groups. Addressing these obligations requires deliberate choices at every stage of the development and deployment pipeline — choices this pilot is designed to make visible and testable within Mercer's own institutional context.

Collectively, these studies also inform the governance strategy adopted in this proposal. Rather than treating ethics and safety as abstract principles, the accompanying risk register operationalises these recommendations through explicit risks, measurable mitigation strategies, and predefined governance triggers. In doing so, the

proposal translates published ethical guidance into practical deployment requirements that can be evaluated throughout the proposed pilot.

3.4 Workflow Integration: Operational Realities Inside the ED

A smaller body of work moves past algorithm development to examine what operational reality looks like once a system is running, and the findings are sobering in useful ways.

Wang et al. [7] studied mobile technology interventions on pre-triage patient flow through direct observation in a live ED. The dominant sources of delay were not clinical but logistical: information handoffs between staff, manual transcription from paper to screen, and documentation bottlenecks that no predictive model resolves because they arise upstream of where the model operates. The single-site design and short observation window limit how broadly the findings travel, but the observational — rather than simulated — methodology gives them real-world weight.

Araouchi and Adda [5] built TriageIntelli, a multimodal prototype that accepted text, image, and structured data inputs simultaneously to produce triage recommendations in lower-resource health centres. Controlled testing showed strong accuracy, but the authors were candid that the system had never been tested in a live clinical environment and that the computational demands of processing all three input types simultaneously may be prohibitive in exactly the settings it was designed for.

El Ariss et al. [10] shifted the question from patient acuity to resource demand, training a machine-learning framework to forecast real-time bed demand and staffing requirements from incoming patient data. Performance on held-out data from the development hospital was promising, but the authors explicitly called for external validation before broader rollout, noting that resource-allocation patterns are tightly coupled to local admission policies — institutional specificity that does not survive transplantation to a different department.

3.5 Validation Gaps: The Evidence We Are Still Missing

The most consistent finding across this body of work is not what AI can do, but what has not yet been properly tested.

Yi, Baik, and Baek [9] conducted a systematic review restricted to prospective studies — the strongest design for evaluating real-world AI triage — and found the evidence base thin. Prospective trials showed that AI-assisted triage can reduce waiting times and improve identification of deteriorating patients, but the small number of qualifying studies and variation in outcome definitions made firm conclusions premature. The review called explicitly for prospective validation in diverse clinical settings before widespread adoption. Porto [8] reached a compatible finding from the algorithm side: even high-performing models were almost exclusively validated retrospectively on the institutions that trained them, a design choice that inflates apparent accuracy and gives no indication of how models perform on a different patient population.

Recent work has begun to address these gaps, though in ways that simultaneously clarify how much remains unresolved. Taylor et al. [11] published a prospective evaluation of a live AI-informed triage decision-support system — implementing a machine-learning ESI recommendation tool at a large academic medical centre and measuring its impact against pre-implementation baselines. AI decision support improved identification of high-acuity patients and reduced undertriage rates in a real clinical environment, providing evidence of a

category Yi et al. [9] had found nearly absent. The critical limitation Taylor et al. [11] acknowledged is that all participating sites were high-income, technologically mature academic centres; transferability to institutions operating under different constraints — Mercer included — remains explicitly unresolved.

Hinson et al. [12], whose equity findings are discussed in Section 3.3, further demonstrated that the mechanism enabling equitable performance across sites — site-specific model calibration — requires deliberate investment that most retrospective single-site studies do not build in. Accuracy and equity performance observed at one institution should not be assumed to transfer without recalibration.

3.6 Synthesis: The Gap This Pilot Addresses

Three things emerge from this body of work with enough consistency to treat as settled. Machine-learning models can predict triage acuity with meaningful accuracy. They fail to achieve adoption when they ignore what nurses need from a tool in a busy department. And the accuracy figures that dominate the published literature are almost universally self-referential — produced by institutions testing their own models on their own historical data.

Recent prospective and multisite studies [11, 12] have begun to challenge this pattern, providing the first substantive evidence of what live-deployment validation looks like and what it can demonstrate. But they also consistently point toward a condition none of them has fully resolved: performance and equity outcomes are shaped by the specific patient population, clinical culture, documentation practices, and institutional infrastructure of the site where a system is deployed. Training-data bias rooted in differential clinical practice across demographic groups [13], workflow friction not captured in simulation [6], and governance gaps that create patient safety risks [14] are institution-specific problems that cannot be resolved by generalising from another department's evidence, however rigorously produced. The guiding principles Chin et al. [15] articulate — representative training data, transparent outputs, clinical primacy, and structured bias audits — provide the ethical framework within which institution-specific deployment must operate.

What does not exist is evidence that an AI triage system, built around the operational constraints of a specific department and validated on that department's own records, can integrate into a live workflow without adding burden or eroding prioritisation quality. That is a question that can only be answered by running the system in the department it is designed for, with the nurses who will use it, on the patients who will be affected by it. That is what this pilot at Mercer General Hospital is designed to do. Recognising that successful deployment depends on more than algorithmic performance alone, the accompanying Risk Analysis translates the principal technical, operational, ethical, and equity risks identified throughout the literature into a structured governance framework with measurable mitigation strategies and ongoing monitoring requirements.



4. Workflow Diagram

Door-to-Disposition Flow — Three AI Plug-in Points

- Patient arrives → Registration → Vital Signs Captured
- AI-1: Symptom Intake Assistant — captures structured presenting complaint data at registration, prior to nurse assessment, to reduce transcription burden and populate model inputs in advance.
- Triage Nurse Assessment → High Acuity? Decision
- AI-2: Triage Recommendation — surfaces as a single-screen overlay within the triage interface during the nurse assessment, displaying recommended ESI level, confidence score, and key driving factors. No keystroke required to dismiss.
- High Acuity path → Immediate Care → Emergency Assessment → Urgent Diagnostics → Critical Treatment
- Standard path → Standard Care → Standard Assessment
- AI-3: Risk Prediction for Waiting Patients — monitors patients in the standard pathway and flags those whose data suggests acuity escalation, prompting re-assessment before deterioration becomes apparent.
- Both paths converge at: Routine/Urgent Diagnostics → Treatment → Patient Reassessment → Disposition Decision (Admit / Discharge / Transfer)

All three AI plug-in points are intentionally designed as passive decision-support tools rather than mandatory workflow checkpoints. This design minimises disruption to existing clinical practice while helping reduce Risk 3 (Alert Fatigue) identified in the accompanying Risk Analysis.

5. Workflow Constraints

Three constraints were identified through observation and staff consultation. Each is named, classified, and described by the mechanism through which it affects AI performance or adoption.

Constraint 1 — Vital Sign Documentation Lag | Technical

During busy periods, nursing staff record vital signs on paper at the bedside and enter them into the electronic health record only when time permits—a gap that typically ranges from 5 to 30 minutes and can exceed one hour during periods of exceptional demand. Because the AI system relies on information contained within the electronic record, recommendations generated before documentation is complete may be based on incomplete clinical information.

Mechanism

Incomplete input data produces an incomplete feature set for the prediction model. Recommendations generated from missing observations carry greater uncertainty and may underestimate patient acuity if the absence of information is not clearly communicated to clinicians.

Design Response

Recommendations generated from incomplete records display explicit data-completeness indicators and identify missing observations before an acuity recommendation is presented. This constraint directly supports mitigation of Risk 2 (Vital-sign Input Incompleteness) by preventing clinicians from unknowingly acting on partially populated records.

Constraint 2 — Nurse Cognitive Load During Peak Demand | Operational

During peak periods—typically weekday evenings and weekend afternoons—triage nurses complete assessments approximately every four to six minutes. Introducing additional screens, mandatory confirmation dialogues, or repetitive interaction steps risks increasing workload and reducing engagement with the system.

Mechanism

Each additional interaction increases cognitive switching and assessment time. If excessive alerts or unnecessary interruptions occur repeatedly, clinicians may begin dismissing recommendations without review, reducing the effectiveness of genuine high-priority alerts.

Design Response

Recommendations appear as passive overlays within the existing triage interface and require no additional keystrokes to dismiss. Alert thresholds will be continuously monitored and adjusted during governance reviews where necessary. This workflow constraint directly supports mitigation of Risk 3 (Alert Fatigue).

Constraint 3 — Dependency on External Reviews and Reports | Structural

Disposition decisions frequently depend on laboratory investigations, diagnostic imaging, or specialist review, all of which may become available well after the initial triage assessment. Recommendations generated before these additional findings are available may no longer reflect the patient's current clinical condition.

Mechanism

The model produces predictions using only the information available at the time of the query. As new clinical information becomes available, the original recommendation may become progressively less representative of the patient's true condition.

Design Response

Every recommendation is timestamped, and clinicians are prompted to request a new assessment whenever significant laboratory, imaging, or specialist information is added to the patient's record. Combined with prospective local validation before deployment beyond Mercer General Hospital, this supports mitigation of Risk 1 (Training-Distribution Shift) by ensuring recommendations remain clinically current and are interpreted within the limitations of the model.

6. Stakeholders

Five clinical stakeholder groups have a direct interest in the pilot's design, conduct, and outcome.

Stakeholder	Top Concern
Triage Nurses	Accurate, explainable recommendations that integrate into the triage interface without adding documentation steps or slowing assessments during peak periods. Nurses must retain clear authority to override the AI output without friction or procedural penalty.
Emergency Physicians	Reliable identification of high-acuity patients before clinical deterioration, with clear audit trails for all AI-assisted decisions. Physicians need confidence that the system's recommendations are based on complete, current data and that disagreements between nurse and algorithm are recorded for review.
Patients	Timely, consistent, and fair assessment regardless of when they arrive or which nurse is on duty. Patients from demographic groups historically subject to under-triage have a specific interest in the pilot's bias audit processes and equity monitoring, as documented by Chang et al. [13] and Hinson et al. [12].
Specialty Clinicians	Appropriate referral prioritisation so that high-acuity patients are escalated before the specialist review queue is reached. Clinicians need confidence that AI-assisted triage does not delay escalation by generating falsely reassuring acuity recommendations.
Hospital Administrators	Measurable reduction in ED overcrowding, demonstrable return on implementation investment, and documented compliance with the ethical and governance obligations of AI deployment, in line with the frameworks set out by Labkoff et al. [14] and Chin et al. [15].

7. Risk Analysis

Successful implementation of AI-assisted emergency department triage depends on more than predictive performance alone. Clinical AI systems operate within complex healthcare environments where technical limitations, workflow pressures, human decision-making, and inequities in historical data can all influence patient outcomes. Consistent with the literature reviewed in Section 3, this proposal adopts a structured risk-management approach that identifies the principal AI-technical, operational, ethical, and equity risks associated with the proposed pilot. Each identified risk includes a likelihood and impact assessment, a practical mitigation strategy, and a measurable signal of success that can be monitored throughout deployment. Rather than treating risk management as a separate governance activity, the controls described below are integrated into the proposed workflow, stakeholder responsibilities, and evaluation strategy.

7.1 Risk Register

#	Risk Name	Category	Likelihood	Impact	Mitigation	Signal of Success
1	Training-distribution shift between Mercer urban ED population and secondary deployment sites — The model is trained exclusively on Mercer's historical case mix. If the Ministry rolls the system out to rural Caribbean hospitals with different disease prevalence (e.g., dengue surge, sickle-cell presentations, higher paediatric volume), the model's calibration may degrade without warning, systematically misclassifying acuity at the new site.	AI-Technical	H	H	Conduct prospective external validation on a held-out dataset from each new deployment site before go-live; compare Level 1–2 sensitivity against the Mercer baseline; halt rollout to that site if the sensitivity gap exceeds 3 percentage points and initiate retraining with local data.	External validation confirms Level 1–2 sensitivity at each new site is within ± 3 percentage points of Mercer training-site performance, verified and documented before each site rollout; site-level performance reviewed quarterly thereafter.

2	Vital-sign input incompleteness producing false-low acuity recommendations during documentation lag periods — Mercer nurses routinely log bedside observations on paper before entering them into the EHR, with a delay of 5–60 minutes (longer during surges). If the model generates an acuity score on a partially empty record, it may produce an artificially low-confidence or low-acuity recommendation that a busy nurse accepts without recognising the missing data.	AI-Technical	H	H	Gate acuity score generation so that if more than two vital sign fields are absent, the system displays only the data-completeness panel and withholds the acuity recommendation until data is logged; timestamp every recommendation with a data-completeness indicator showing which fields were present at the moment of generation.	Weekly EHR audit confirms zero acuity scores were generated with more than two vital-sign fields missing; data-completeness flag displayed on 100% of recommendations throughout the audit period.
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3	Alert fatigue from sustained high false-positive rate causing triage nurses to systematically dismiss AI escalation flags during peak-volume shifts — At peak demand, nurses complete assessments every four to six minutes. If the system fires frequently on non-critical patients, nurses will habituate to dismissing alerts — including those flagging genuine Level 1 cases. The passive-overlay design reduces friction but does not eliminate the risk of systematic inattention.	Operational	H	H	Set the alert-triggering threshold before deployment so positive predictive value for high-acuity flags reaches $\geq 40\%$; monitor alert-to-clinical-action rate weekly and report findings to the named clinical owner; adjust the threshold if the rate falls consistently below 50%.	Alert-to-clinical-action rate remains $\geq 50\%$ across all monitored shifts; any two consecutive weeks below 40% trigger a mandatory clinical governance review with documented threshold adjustment within seven working days.
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4	EHR integration failure leaving triage nurses without AI support and without a tested manual fallback protocol — The system reads directly from the EHR to pull vital signs and presenting complaints. Planned maintenance windows, unplanned outages, or connectivity failures will interrupt this feed. Nurses who have become accustomed to AI-assisted triage may be slower or less consistent when the tool is unavailable if no manual fallback has been practised.	Operational	L	M	Design and document a manual triage protocol for use during AI system downtime before go-live; train all triage-qualified staff on both AI-assisted and manual workflows as part of pre-deployment competency assessment; conduct quarterly downtime simulation drills.	100% of triage-qualified staff complete the manual fallback drill before go-live and annually thereafter; drill completion records audited by the clinical governance lead each quarter; drill performance reviewed at governance meetings.
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5	Automation bias causing triage nurses to accept AI acuity recommendations without independent clinical assessment, particularly during high cognitive-load periods — Evidence reviewed in the proposal (Da'Costa et al., 2025) identifies clinician deference to opaque AI tools as a primary adoption and safety risk. Even with visible reasoning, the passive-overlay design may anchor the nurse's assessment, particularly when the department is under pressure and the marginal cognitive cost of independent evaluation is highest.	Ethical	M	H	Embed override-documentation into pre-deployment nurse training; require staff to complete structured simulation scenarios in which the AI recommendation is deliberately set to the wrong acuity level and must be independently identified and overridden; monitor live override rates weekly against a pre-defined expected range.	Staff correctly override AI recommendations in simulation scenarios in ≥90% of cases at 6-month and 12-month post-deployment assessments; live override rate (nurse assessment differing from AI score) remains within 5–20%; a sustained drop below 5% in any three-week window triggers a mandatory training review.
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6	Absence of patient disclosure that AI contributed to their triage prioritisation, creating an informed consent gap — The proposal does not include a patient-facing disclosure mechanism. Patients at Mercer General Hospital's ED will be triaged with AI input without being informed that an algorithmic tool contributed to their acuity assignment. This is inconsistent with WHO transparency and autonomy principles and may conflict with emerging Caribbean data governance standards.	Ethical	M	M	Draft a plain-language patient-facing disclosure statement explaining that an AI decision-support tool assists triage nurses; have the statement reviewed by a patient advocacy group and legal counsel before deployment; include disclosure in admissions registration documentation and at the triage desk.	Patient-facing disclosure language signed off by patient advocate group and legal counsel before go-live; patient satisfaction survey at 3-month post-deployment review includes one AI-disclosure comprehension question; comprehension rate ≥70% recorded.
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7	Systematic under-triage of elderly patients (>65 years) with atypical high-acuity presentations if this demographic is underrepresented in Mercer's historical training data — Older patients frequently present with atypical symptoms for conditions such as myocardial infarction, sepsis, and stroke (e.g., absence of fever, vague pain, altered cognition without clear cause). If patients >65 are underrepresented in the Mercer training cohort, the model may fail to flag them as high acuity — an error with direct mortality risk.	Equity	M	H	Audit training dataset demographics before model training; if patients >65 represent fewer than 15% of training cases, oversample from Mercer records or supplement with de-identified retrospective data from comparable Caribbean hospitals; disaggregate sensitivity and specificity by age group in all validation reports.	Quarterly subgroup performance audit confirms Level 1–2 sensitivity for patients >65 is within 5 percentage points of overall model sensitivity; any sustained gap >5 percentage points triggers a mandatory retraining review; results published in clinical governance reports.
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8	Infrastructure disparity if Ministry rollout reaches rural or lower-resource Caribbean hospitals without stable EHR connectivity or compatible hardware — Mercer is an urban, well-resourced hospital. Rural sites with intermittent power, older hardware, or unreliable internet connectivity may experience frequent AI system failures that Mercer's environment would not reveal during development. This creates an inequitable two-tier service in which urban patients receive AI-assisted triage while rural patients, who often have fewer healthcare alternatives, do not.	Equity	M	M	Develop a mandatory site readiness checklist covering EHR compatibility, hardware specifications, connectivity reliability, and power backup; require sign-off from the IT lead and named clinical owner at each site before deployment; document an offline fallback procedure for sites with intermittent connectivity.	Site readiness checklist completed and signed off before each rollout; monthly uptime monitoring confirms ≥98% AI system availability at each deployed site; any site falling below threshold receives a documented infrastructure improvement plan within 30 days.
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7.2 Risk Memo

Although all eight risks require active management, three represent the greatest potential threat to patient safety because they combine a high likelihood of occurrence with potentially serious clinical consequences.

The first is **training-distribution shift (Risk 1)**. Much of the literature reviewed in this proposal demonstrates that models performing well within their development institution frequently experience degraded performance when applied elsewhere. Differences in patient demographics, disease prevalence, documentation practices, and clinical workflows may substantially alter model calibration without obvious warning. For an emergency department triage system, even relatively small reductions in sensitivity for high-acuity patients may lead to inappropriate prioritisation and delayed treatment. For this reason, prospective external validation using site-specific data is proposed before any deployment beyond Mercer General Hospital. Nevertheless, residual uncertainty remains because successful validation at one additional institution cannot guarantee equivalent performance across all future healthcare settings. Continued monitoring and periodic recalibration will therefore remain necessary throughout the system's operational life.

A second high-priority concern is **alert fatigue (Risk 3)**. Clinical decision-support systems only improve patient outcomes if healthcare professionals continue to engage with their recommendations. Excessive false-positive alerts encourage clinicians to disregard notifications, increasing the likelihood that genuinely deteriorating patients may subsequently be overlooked. The proposed passive-overlay interface reduces unnecessary interruption, while routine monitoring of alert-to-clinical-action rates provides an objective measure of whether recommendations continue to influence clinical behaviour appropriately. Governance thresholds requiring formal review if clinician engagement declines ensure that deterioration in system performance is recognised before patient safety is compromised. Even so, human behaviour remains inherently variable, particularly during periods of exceptional workload, meaning that alert fatigue cannot be eliminated entirely.

The third priority is **automation bias (Risk 5)**. AI recommendations may unintentionally anchor clinical judgement, particularly when staff are working under significant cognitive pressure. Although explainable outputs and clinician override mechanisms reduce this risk, they cannot fully prevent over-reliance on algorithmic advice. The proposal therefore combines transparent recommendations with structured simulation training in which clinicians must deliberately recognise and reject incorrect AI outputs. Monitoring live override rates further provides an early indicator of inappropriate reliance on the system. Nevertheless, professional judgement remains central to safe emergency care, and no technical safeguard can entirely remove the possibility of inappropriate deference to AI recommendations. Continuous education, governance oversight, and reinforcement of clinician accountability therefore remain essential throughout the pilot.

Residual Uncertainty

Despite the comprehensive mitigation strategies proposed, uncertainty cannot be completely eliminated. External validation improves confidence that the model generalises beyond the Mercer population but does not guarantee equivalent performance across different hospitals, patient populations, or healthcare infrastructures. Similarly, retrospective evaluation using historical records cannot fully replicate the complexity of live clinical practice, where documentation behaviour, workload, and disease prevalence may change over time. Consequently, successful completion of this 12-week pilot should be viewed as evidence supporting further prospective multicentre evaluation rather than definitive proof of safety or effectiveness. Ongoing governance, continuous performance monitoring, and periodic model recalibration will remain necessary before any wider implementation is considered.

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